2-mavzu. Determinantlar nazariyasi.

Reja:

2.1. Ikkinchi va uchunchi tartibli determinantlar

2.2. Minor va algebraik to'ldiruvchi

2.3. Yuqori tartibli detirminantlarni hisoblash

Tayanch iboralar: *determinant, tartib, uchburchak usuli, yoyilma.*

2.1. Ikkinchi va uchunchi tartibli determinantlar.

2 tartibli (2×2 o'lchovli) matritsaning determinanti deb qarama-qarshi burchakdagi elementlar ko'paytmasining ayirishdan hosil bo'lgan songa aytiladi. Odatda determinantlarni matritsalardan farqlash uchun ikki tomonidan vertikal to'g'ri chiziqlar bilan belgilanadi, matritsa kabi kvadrat qavslar emas.

2×2 o'lchovli A matritsa determinant |**A**| kabi belgilanadi.

Shuning uchun

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

Misol.

$A = \begin{bmatrix} 5 & 7 \\ 4 & 9 \end{bmatrix}$ matritsa determinanti topilsin.

Yechish. Yuqoridagi formuladan foydalanib hisoblaymiz

$$|A| = \begin{vmatrix} 5 & 7 \\ 4 & 9 \end{vmatrix} = 5 \times 9 - 7 \times 4 = 45 - 28 = 17$$

Agar matritsa satrlari yoki ustunlari orasida chiziqli bog'lanish mavjud bo'lsa, uning determinant nolga teng bo'ladi va u xos matritsa deyiladi.

Masalan, $A = \begin{bmatrix} 5 & 8 \\ 10 & 16 \end{bmatrix}$ matritsa determinanti

$$|A| = \begin{vmatrix} 5 & 7 \\ 4 & 9 \end{vmatrix} = 5 \times 16 - 8 \times 10 = 80 - 80 = 0,$$

shunig uchu A matritsa xos matritsadir.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

uchinchi tartibli martisa uchun |**A**| determinant quyidagicha hisoblanadi:

$$|A| = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Bu birinchi satr elementlarini shu element turgan satr va ustunni o'chirishdan hosil bo'lgan matritsa determinantiga ko'paytirishdan hosil qilinmoqda. Masalan,

3x3 o'lchovli berilgan matritsa 1 satr va 1 ustunu o'chirilishidan hosil bo'lgan determinant elementga ko'paytirilgan. Agar biz a_{11} belgilashga e'tibor bersak, u holda bu usulni satr bo'yicha qo'llaganda ishora almashadi. Demak, ikkinchi qo'shiluvchi manfiy ishorali bo'ladi.

Misol.

$A=\begin{bmatrix} 4 & 6 & 1 \\ 2 & 5 & 2 \\ 9 & 0 & 4 \end{bmatrix}$ matritsa determinanti topilsin.

Yechish. Yuqoridagi formulaga ko'ra, birinchi satr elementlari bo'yicha yoyilma quyidagicha bo'ladi

$$|A|=4\begin{vmatrix} 5 & 2 \\ 0 & 4 \end{vmatrix}-6\begin{vmatrix} 2 & 2 \\ 9 & 4 \end{vmatrix}+\begin{vmatrix} 2 & 5 \\ 9 & 0 \end{vmatrix}=$$

$$=4(20-0)-6(8-18)+(0-45)=80+60-45=95$$

Ta'rif. *n-tartibli kvadrat* $A=(a_{ij})$ *matritsaning determinanti deb, quyidagi tenglik bilan aniqlangan songa aytiladi:*

$$|A|=\sum_{k=1}^{n} a_{1k}\ A_{1k}$$

Bu ta'rifdan foydalanib 2 va 3 tartibli determinantlarni hisoblash uchun quyidagi formulalarni hosil qilamiz:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}=\sum_{k=1}^{2} a_{1k}A_{1k}=a_{11}A_{11}+a_{12}A_{12}=a_{11}a_{22}-a_{12}a_{21}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}=\sum_{k=1}^{3} a_{1k}A_{1k}=a_{11}A_{11}+a_{12}A_{12}+a_{13}A_{13}=a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}-a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}+a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}=$$

$$=a_{11}(a_{22}a_{33}-a_{23}a_{32})-a_{12}(a_{21}a_{33}-a_{23}a_{31})+a_{13}(a_{21}a_{32}-a_{22}a_{31})=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+$$

$$+a_{21}a_{32}a_{13}-a_{13}a_{22}a_{31}-a_{32}a_{23}a_{11}-a_{21}a_{12}a_{33}$$

Determinant xossalari

1-xossa. *Agar* A*-matritsaning biron-bir satridagi (ustunidagi) barcha elementlari nolga teng bo'lsa, u holda uning determinanti nolga teng bo 'ladi.*

2-xossa. *Agar* A*-matritsaning biron-bir satr (ustun) elementi* λ *soniga ko'paytirilsa, determinant qiymati ham* λ *soniga ko'payadi, ya'ni* $\lambda\cdot|A|$ *ga teng bo 'ladi.*

3-xossa. A*-matritsa va uning transponirlangani* A' *matritsalarning determinantlari teng bo'ladi, ya'ni* $|A|=|A'|$ *tenglik o'rinlidir.*

4-xossa. *Agar* A*- matritsaning ikkita qo'shni satrlari o'rnini almashtirsak, hosil bo'lgan yangi* A_1 *matritsaning determinanti* A*-matritsa determinantining teskari ishora bilan olinganiga teng bo 'ladi, ya'ni* $|A_1|=-|A|$ *tenglik o'rinli bo'ladi.*

5-xossa. *Agar* A*-matritsa bir xil ikki satrga (ustunga) ega bo'lsa, u holda uning determinanti nolga teng , ya'ni* $|A| = 0$ *bo 'ladi.*

6-xossa. *Agar* A*-matritsada ikki satrning (ustun) mos elementlari proporsional bo'lsa, u holda uning determinanti nolga teng, ya'ni* $|A| = 0$ *bo 'ladi.*

7-xossa. *Agar* A *matritsaning biron satr (ustun) elementlarini boshqa satr (ustun) mos elementlarining algebraik to'ldiruvchilariga ko'paytirib yig'indi hosil qilsak, bunday yig'indi nolga teng bo'ladi, ya'ni* $\sum_{k=1}^{n} a_{ik} A_{jk} = 0, \quad i \neq j$.

8- xossa. A *matritsaning biron-bir satri (ustuni) elementlarini bir xil songa ko'paytirib, boshqasiga qo'shishdan hosil bo'lgan* A_1 *- matritsaning determinanti* A *matritsa determinantiga teng bo'ladi, ya'ni* $|A_1| = |A|$.

9-xossa. $b_1, \ b_2, \ldots, b_n$ *sonlarni n-tartibli* A *matritsaning berilgan satr (ustun) mos elementlarining algebraik to'ldiruvchilariga ko'paytmasining yig'indisi,* A *matritsaning berilgan satr (ustun) elementlarining* $b_1, \ b_2, \ldots, b_n$ *sonlari bilan almashtirilgan matritsa determinantiga teng bo 'ladi.*

10-xossa. *n-tartibli kvadrat* A *va* B *matritsalar uchun* $|A \cdot B| = |A| \cdot |B|$ *tenglik o'rinli bo'ladi, ya'ni matritsalar ko'paytmasining determinanti, ularning determinantlari ko'paytmasiga teng bo'ladi.*

Misol. $\begin{vmatrix} 1 & 2 & 1 \\ 3 & -5 & 3 \\ 2 & 7 & -1 \end{vmatrix} = ?$

Yechish. $\begin{vmatrix} 1 & 2 & 1 \\ 3 & -5 & 3 \\ 2 & 7 & -1 \end{vmatrix} = 5 + 12 + 21 + 10 - 21 + 6 = 33.$

2.2. Minor va algebraik to'ldiruvchi

Laplas yoyilmasi yordamida har qanday tartibli determinantni hisoblash mumkin. Lekin buning uchun ba'zi tushunchalar bilan tanishishimiz kerak (ba'zilaridan biz foydalandik ham).

Minorlar. A matritsa |**M*ij*** | minori deb ***i*** satr va ***j*** ustunni o'chirishdan hosil bo'lgan determinantga aytiladi.

Masalan,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

matritsa uchun

$$|M_{11}| = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$$

Misol.

$A = \begin{bmatrix} 8 & 2 & 3 \\ 1 & 9 & 4 \\ 4 & 3 & 6 \end{bmatrix}$ matritsa uchun $|M_{31}|$ minorni hisoblang.

Yechish. Uchinchi satr va birinchi ustunni o'chirib

$$|M_{31}| = \begin{vmatrix} 2 & 3 \\ 9 & 4 \end{vmatrix} = 8 - 27 = -19$$

minorni hosil qilamiz.

Minor ta'rifidan foydalanib, 3 tartibli determinant hisoblash formulasini quyidagicha yozish mumkin

$$|A| = a_{11}|M_{11}| - a_{12}|M_{12}| + a_{13}|M_{13}|$$

Algebraik to'ldiruvchi. $(-1)^{i+j}$ ishora aniqlikdagi |**M***ij* | minorga $|C_{ij}|$ algebraik to'ldiruvchi deyiladi. $|C_{ij}|$ algebraik to'ldiruvchi ishorasi $(-1)^{i+j}$. Shunday qilib, satr va ustun nomerlari yig'indisi toq bo'lsa,u holda ishora manfiy bo'ladi. Masalan, 3 tartibli **A** matritsada $|C_{12}|$ algebraik to'ldiruvchini topish uchun 1 satr va 2 ustunni o'chiramiz, $i + j = 3$ bo'lgani uchun hosil bo'lgan determinantni $(-1)^3$ ga ko'paytiramiz. Demak,

$$|C_{12}| = (-1)^3 \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = (-1) \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$$

Misol.

$A = \begin{bmatrix} 8 & 2 & 3 \\ 1 & 9 & 4 \\ 4 & 3 & 6 \end{bmatrix}$ matritsa uchun $|C_{22}|$algebraik to'ldiruvchini toping.

Yechish. $i + j = 4$ bo'lgani uchun

$$|C_{22}| = (-1)^4 \begin{vmatrix} 8 & 3 \\ 4 & 6 \end{vmatrix} = (+1)(48 - 12) = 36$$

Algebraik to'ldiruvchi ta'rifidan foydalanib, 3 tartibli determinant hisoblash formulasini quyidagicha yozish mumkin

$$|A| = a_{11}|C_{11}| + a_{12}|C_{12}| + a_{13}|C_{13}| \qquad (1)$$

Bu munosabat xuddi A determinantni hisoblash formulasidek ko'ringani bilan, shuni ta'kidlash kerakki, bu yerdagi ikkinchisi element ishorasi musbatligiga qaramasdan unga mos algebraik to'ldiruvchilardan manfiydir.

2.3. Yuqori tartibli detirminantlarni hisoblash

Laplas yoyilmasi. Har qanday *n* tartibli matritsa determinantini Laplas yoyilmasidan foydalanib hisoblash mumkin

$$|A| = \sum_{i=1}^{n} a_{ij} |C_{ij}| \ , \quad j = 1,2,3,...,n$$

$$|A| = \sum_{j=1}^{n} a_{ij} |C_{ij}| \ , \ i = 1,2,3,...,n$$

bunda yig'indi 1 dan n gacha ustun bo'yicha ham (j) yoki satr bo'yicha ham (i) bo'lishi mumkin. Agar siz (1) 3 tartibli determinant hisoblash formulasiga e'tibor bersangiz, u Laplas yoyilmasini ifodalaganini anglaysiz. Agar berilgan matritsa 4 yoki yuqori tartibli bo'lsa, u holda algebraik to'ldiruvchilar 3 tartibli yoki yuqori tartibli bo'ladi. shunday qilib, Laplas yoilmasi tarbni pasytirish orqali hisoblashdir, ya'ni berilgan determinantning tartibi bittaga pasayadi. Har qanday yuqori tartibli determinantni Laplas yoyilmasidan foydalanib hisoblaganda unu tartibini ikkinchi tartibgacha tushirib hishoblash qulay.

Determinant tartibi yetarli katta bo'lganda ko'p hisoblashlar bajarishga to'g'ri keladi va shuning uchun tezroq hisoblash usullari zarur bo'ladi.

Oldin bu usulda hisoblashga misol ko'raylik.

Misol. Laplas yoyilmasidan foydalanib, quyidagi matritsa determinantni hisoblang

$$A = \begin{bmatrix} 8 & 10 & 2 & 3 \\ 0 & 5 & 7 & 10 \\ 2 & 2 & 1 & 4 \\ 3 & 4 & 4 & 0 \end{bmatrix}$$

Yechish. Matritsa determinantini hisoblash uchun uni birinchi ustun elementlari bo'yicha yoyamiz(bu ustonda nol elementi bo'lgani uchun hisoblashda bitta kam uchinchi tartibli determinant hisoblash qulaylik tug'diradi)

$$|A| = 8\begin{vmatrix} 5 & 7 & 10 \\ 2 & 1 & 4 \\ 4 & 4 & 0 \end{vmatrix} - 0\begin{vmatrix} 10 & 2 & 3 \\ 2 & 1 & 4 \\ 4 & 4 & 0 \end{vmatrix} + 2\begin{vmatrix} 10 & 2 & 3 \\ 5 & 7 & 10 \\ 4 & 4 & 0 \end{vmatrix} - 3\begin{vmatrix} 10 & 2 & 3 \\ 5 & 7 & 10 \\ 2 & 1 & 4 \end{vmatrix}$$

Endi uchinchi tartibli har bir determinantni yana birinchi ustun elementlari bo'yicha yoyamiz:

$$|A| = 8\left(5\begin{vmatrix}1 & 4\\4 & 0\end{vmatrix} - 2\begin{vmatrix}7 & 10\\4 & 0\end{vmatrix} + 4\begin{vmatrix}7 & 10\\1 & 4\end{vmatrix}\right) + 2\left(10\begin{vmatrix}7 & 10\\4 & 0\end{vmatrix} - 5\begin{vmatrix}2 & 3\\4 & 0\end{vmatrix} + \begin{vmatrix}2 & 3\\7 & 10\end{vmatrix}\right) -$$

$$-3\left(10\begin{vmatrix}7 & 10\\1 & 4\end{vmatrix} - 5\begin{vmatrix}2 & 3\\1 & 4\end{vmatrix} + 2\begin{vmatrix}2 & 3\\7 & 10\end{vmatrix}\right) =$$

$$= 8\left[5(-16) - 2(-40) + 4(18)\right] + 2\left[10(-40) - 5(-12) + 4(-1)\right] -$$

$$-3\left[10(18) - 5(5) + 2(-1)\right] =$$

$$= 8[-80 + 80 + 72] + 2[-400 + 60 - 4] - 3[180 - 25 - 2] =$$

$$= 8(72) + 2(-344) - 3(153) =$$

$$= 576 - 688 - 459 = -571$$

Ta'rif. *n-tartibli kvadrat $A = (a_{ij})$ matritsa a_{ij} elementining M_{ij}-minori deb, A-matritsaning i-satri va j-ustunini o'chirishdan keyin hosil bo'lgan $(n-1)$ tartibli matritsa determinantiga aytiladi.*

Ta'rif. *n-tartibli $A = (a_{ij})$ matritsa a_{ij}-elementining algebraik to'ldiruvchisi A_{ij} - deb quyidagi songa aytiladi $A_{ij} = (-1)^{i+j} M_{ij}$.*

Yig'indi $\sum_{s=1}^{n} a_{is} A_{is}$ – i-satr bo'yicha yoyilma, $\sum_{k=1}^{n} a_{kj} A_{kj}$ yig'indi esa, j-ustun bo'yicha yoyilma deb ataladi.

Ta'rif. *n-tartibli kvadrat $A = (a_{ij})$ matritsaning determinanti deb, quyidagi tenglik bilan aniqlangan songa aytiladi:*

$$|A| = \sum_{k=1}^{n} a_{1k} A_{1k}$$

Teorema (Laplas teoremasi). *Istalgan i va j lar uchun*

$$\sum_{s=1}^{n} a_{is} A_{is} = \sum_{k=1}^{n} a_{kj} A_{kj} = |A| \quad i = \overline{1,n}; \quad j = \overline{1,n}$$

tenglik o'rinli bo 'ladi.

Misol. $\begin{vmatrix}2 & -1 & 3 & 0\\-5 & 3 & 1 & -2\\-3 & -2 & 0 & 4\\1 & 5 & -4 & 2\end{vmatrix}$ determinant hisoblansin.

I usul. Dastlab, to'rtinchi satr elementlari bo'yicha yoyib hisoblaymiz

$$|A| = a_{41}A_{41} + a_{42}A_{42} + a_{43}A_{43} + a_{44}A_{44} = -a_{41}M_{41} + a_{42}M_{42} - a_{43}M_{43} + a_{44}M_{44}.$$

$$|A| = \begin{vmatrix}2 & -1 & 3 & 0\\-5 & 3 & 1 & -2\\-3 & -2 & 0 & 4\\1 & 5 & -4 & 2\end{vmatrix} = -\begin{vmatrix}-1 & 3 & 0\\3 & 1 & -2\\-2 & 0 & 4\end{vmatrix} + 5\begin{vmatrix}2 & 3 & 0\\-5 & 1 & -2\\-3 & 0 & 4\end{vmatrix} + 4\begin{vmatrix}2 & -1 & 0\\-5 & 3 & -2\\-3 & -2 & 4\end{vmatrix} + 2\begin{vmatrix}2 & -1 & 3\\-5 & 3 & 1\\-3 & -2 & 0\end{vmatrix} = 546.$$

II usul. Endi, determinantning xossalaridan foydalanib, uchinchi ustun elementlarini nolga aylantiramiz va shu ustun bo'yicha yoyib hisoblaymiz:

$$\begin{vmatrix} 2 & -1 & 3 & 0 \\ -5 & 3 & 1 & -2 \\ -3 & -2 & 0 & 4 \\ 1 & 5 & -4 & 2 \end{vmatrix} = \begin{vmatrix} 17 & -10 & 0 & 6 \\ -5 & 3 & 1 & -2 \\ -3 & -2 & 0 & 4 \\ -19 & 17 & 0 & -6 \end{vmatrix} = -\begin{vmatrix} 17 & -10 & 6 \\ -3 & -2 & 4 \\ -19 & 17 & -6 \end{vmatrix} = -\begin{vmatrix} 17 & -10 & 6 \\ -3 & -2 & 4 \\ -2 & 7 & 0 \end{vmatrix} =$$

$$= 2\begin{vmatrix} -10 & 6 \\ -2 & 4 \end{vmatrix} + 7\begin{vmatrix} 17 & 6 \\ -3 & 4 \end{vmatrix} = 2(-40+12) + 7(68+18) = 546.$$

Bu usulni “*determinantni tartibini pasaytirib hisoblash usuli*” deb ham yuritiladi.

Mavzuni mustaxkamlash test savollari.

1. Ikkinchi tartibli determinant qanday aniqlanadi?

A) $a_{11}a_{22}+a_{12}a_{21}$

B) $a_{11}a_{22}-a_{12}a_{21}$

C) $a_{11}+a_{22}$

D) $a_{11}-a_{22}$

To‘g‘ri javob: B

2. Uchunchi tartibli determinantni Sarrus qoidasi orqali hisoblash qachon qo‘llaniladi?

A) Faqat simmetrik matritsada

B) Faqat nolli elementli matritsada

C) Faqat 2×2 matritsada

D) Faqat 3×3 matritsada

To‘g‘ri javob: D

3. **Quyidagilardan qaysi biri uchunchi tartibli determinantning umumiy ko‘rinishi?**

A) $a_{11}a_{22}a_{33} - a_{13}a_{22}a_{31}$

B) $a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}$

C) $a_{11} + a_{22} + a_{33}$

D) $a_{11}a_{12}a_{13}$

To‘g‘ri javob: B

4. Minor nima?

A) Matritsaning detirminanti

B) Matritsadagi eng kichik son

C) Elementga tegishli kichik determinant, u o‘sha element joylashgan satr va ustunni o‘chirib tashlab hisoblanadi

D) 0 ga teng element

To‘g‘ri javob: C

5. Algebraik to‘ldiruvchi qanday aniqlanadi?

A) Minor ustiga "+" belgisi qo‘yiladi
B) Minorni ko‘paytma bilan almashtirish
C) Minorni *(−1)i+j(-1)^{i+j}(−1)i+j* ga ko‘paytirish orqali
D) Minorni determinant bilan qo‘shish orqali
To‘g‘ri javob: C

6. Yuqori tartibli determinantlarni hisoblashning asosiy usuli bu –
A) Minorlar bo‘yicha yoyish usuli
B) Qavslar qoidasi
C) Matritsani transpozitsiya qilish
D) Faol ustunni ko‘paytirish
To‘g‘ri javob: A

7. Minorlar bo‘yicha yoyishda qanday qator yoki ustun tanlanadi?
A) Har doim birinchi qator
B) Faqat so‘nggi ustun
C) O‘zimiz istagan qator yoki ustun
D) Faqat diagonal
To‘g‘ri javob: C

8. Agar determinantning biror satri yoki ustuni nollar bilan to‘ldirilgan bo‘lsa, determinant natijasi qanday bo‘ladi?
A) 1 ga teng
B) -1 ga teng
C) O‘zgarmaydi
D) 0 ga teng
To‘g‘ri javob: D

9. Kvadrat matritsaning determinanti 0 bo‘lsa, bu nimani bildiradi?
A) Matritsa diagonal
B) Matritsa teskari matritsaga ega emas
C) Matritsa nollardan iborat
D) Matritsa tenglamasi yo‘q
To‘g‘ri javob: B

10. Quyidagi determinantni hisoblang:

$$\begin{vmatrix} 2 & 0 \\ 1 & 3 \end{vmatrix}$$

A) 6
B) 5
C) 2
D) -6
To‘g‘ri javob: A (Chunki: 2·3−0·1=62 \cdot 3 - 0 \cdot 1 = 62·3−0·1=6)